

SHOWCASE

Sarcos Robotics Guardian XO

Sarcos Robotics has created a powered exosuit that when worn will allow its users to lift about 100 kilos like it's nothing. Exosuits or power suits have been a mainstay in science fiction for decades. The idea of wearing a mechanically-powered suit that significantly boosts your physical capabilities has been a dream of humanities for a long time. In the recent years, we've been getting closer and closer to actually accomplishing that dream. We're still quite a ways from actually having an Iron Man suit though.



According to Sarcos, the Guardian XO focuses on not impeding its wearer's natural range of motion. Previous iterations of exosuits were usually a bit bulky and harder to control and steer.

The XO has no wires and runs on electricity with a runtime of up to two hours. You can, however, stock up on battery packs to keep the suit running for however long you want.

Wearing the Guardian XO also takes close to no time, as per the company, and is also supposedly very easy for new users to pick up.

The Guardian XO robot exosuit is expected to go on sale sometime in Q3 of 2020, and alpha units will ship for approximately \$100,000 per unit.



NASA NEID

The NASA and NSF funded NEID or the NN-EXPLORE Exoplanet Investigations with Doppler spectroscopy, is an instrument that's mounted to the WIYN telescope at the Kitt Peak National Observatory.

The NEID is different from conventional planet-hunting telescopes. Where traditional planet hunting telescopes usually discover planets by looking at periodic dips in light coming from stars, the NEID is looking to actually weigh them.

It does this by looking for a "wobble" in stars. This wobble usually only occurs when a star has a planet orbiting it. Larger planets cause a bigger wobble. Planets closer to stars also cause bigger wobbles. In our own solar system for example, Jupiter causes the biggest wobble, making our Sun move back and forth at roughly 43 feet/second. The NEID is made specifically to detect this wobble. While other instruments can detect a wobble of up to 3 feet per second, the NEID is tuned to detect wobbles at three times the precision. This means that it's able to detect smaller planets around smaller stars, giving it a greater chance to discover planets similar to Earth in size.

Another interesting thing about this method of discovery is the fact that it gives researchers more information about the planets, such as whether it's rocky or gassy, its surface temperature, and its dimensions.

Augmedics Xvision

Augmented Reality right now is normally associated with gaming, and even with furniture apps, to ensure that the table you're buying fits your room's dimensions accurately. But AR has taken actually begun taking strides in other fields as well, and one of them is the medical field. The Augmedics Xvision is a head-mounted AR display which uses augmented reality to assist with surgical practices. Using what's been dubbed the Xvision System, this is an augmented reality image guidance system, which can help guide surgeons during operations. Xvision has



also received approval from the FDA for use in actual surgeries. The system allows surgeons to visualise a patient's anatomy in 3D while operating. The head-mounted display allows them to look inside patients, seeing through skin as though they had X-Ray vision. The headset is lightweight and built so as to not impede surgeons in any way.

It's purely meant to guide them, by making it as easy as possible to access information about the patient. Furthermore, Augmedics have added instruments to allow their display to navigate through medical instruments patients might have on them, such as pacemakers, or even implants.